

QuIP: Experimental Design for Expensive Simulators with Many Qualitative Factors via Integer Programming

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Joint work with Simon Mak

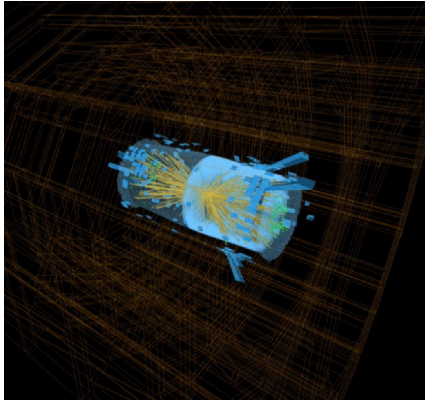
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September 21st, 2026



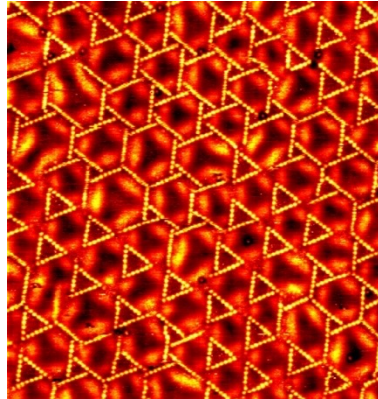
"How hard can
doing a PhD be?"
– Sheldon, 2026

Computer experiments

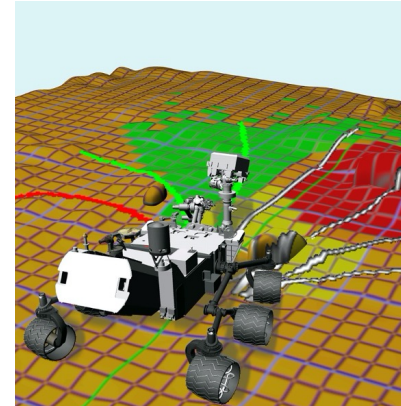


Particle physics

Liyanage et al. 2022, Ji et al.
2024 (a,b), Li et al. 2026



Nanomaterial
optimization



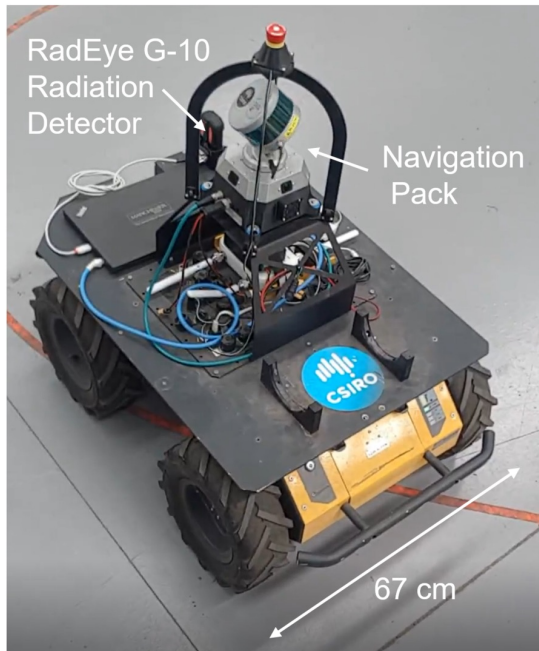
Path planning

- Virtual representation of **complex** phenomena via **computer simulations**
- Proxies for **physical** experiments (**expensive** and/or **infeasible**)

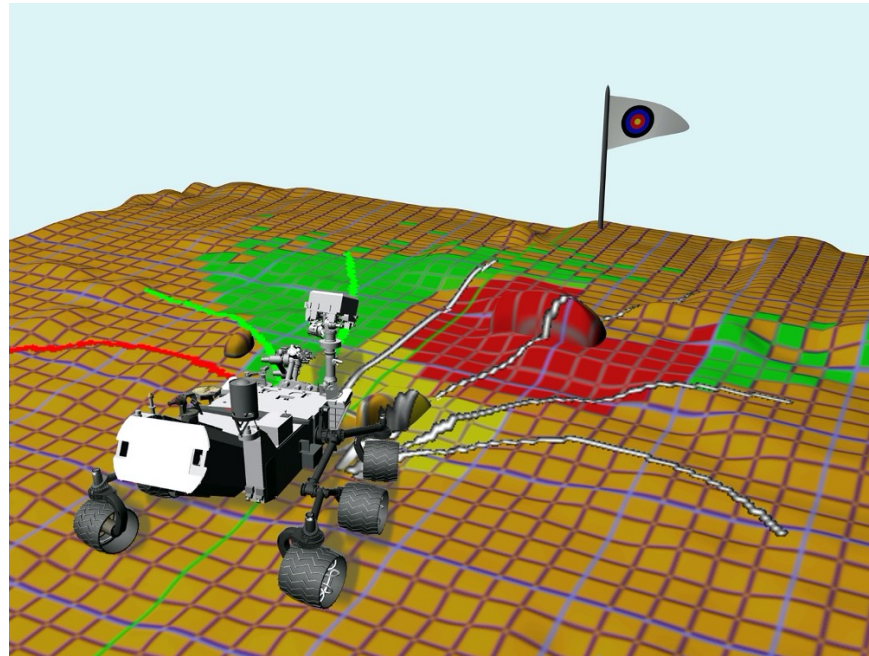
Path planning

Experimental design can be **high-dimensional** in scientific applications:

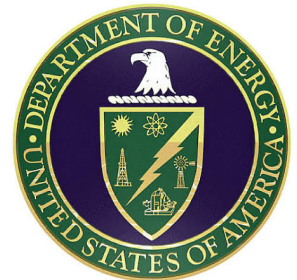
- **Path planning** of **robotic** vehicles for **radiation mapping**
- **Goal**: Find the **quickest navigation path** that takes the robot to destination



Groves et al. (2021)

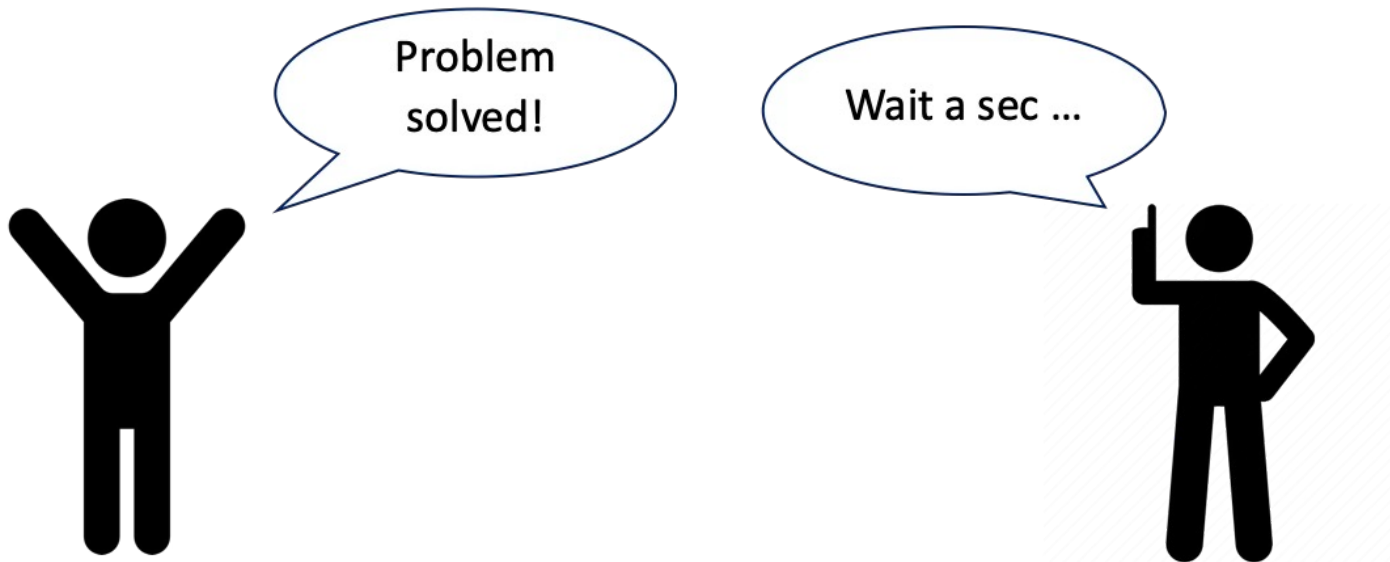


<https://www-robotics.jpl.nasa.gov/>



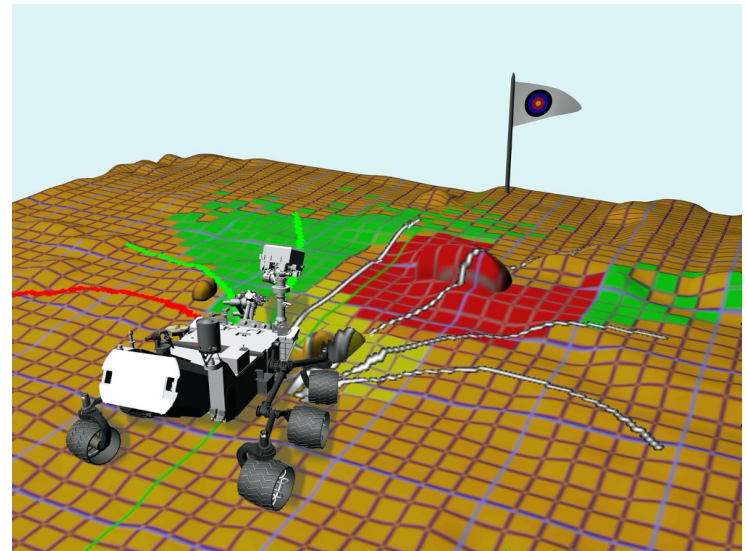
Traditional Approaches

- Dijkstra's algorithm and its variants (Dijkstra, 1976)
 - Particle swarm optimization (Kennedy and Eberhart, 1995)
 - Rapidly exploring random trees (LaValle and Kuffner, 2001)
- and so on



Challenges

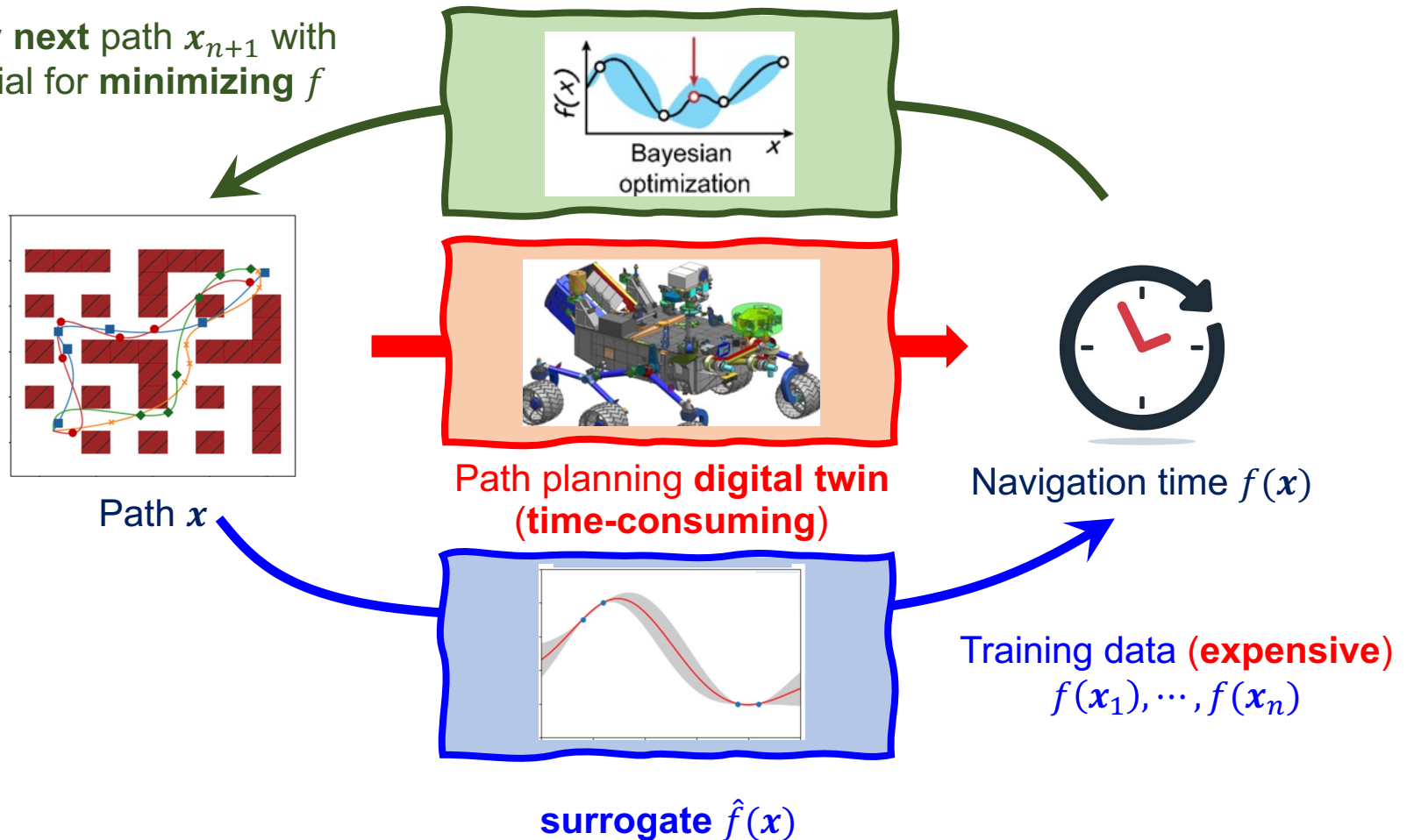
- **Path planning** is now needed for **highly sophisticated** systems.
- **Physical** experiments are **infeasible**.
- Path planning using a **high-fidelity simulator/digital twin**:
 - For a given path x , **simulate** its navigation time $f(x)$ over the terrain
 - **Costly multi-physics** simulations
 - $\mathcal{O}(\text{minutes})$ / path
 - Hinders **real-time** navigation



<https://www-robotics.jpl.nasa.gov/>

Path planning

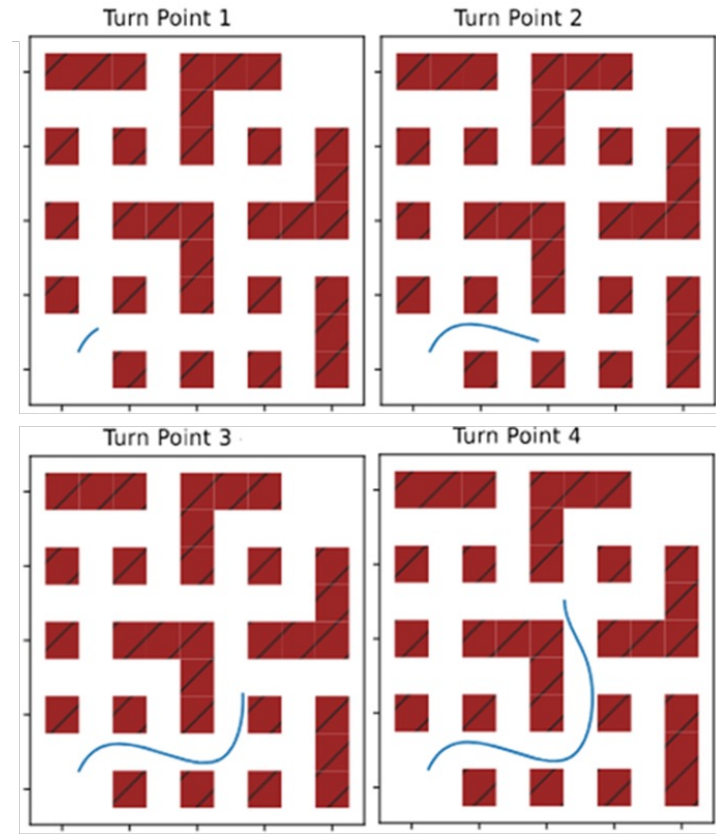
Identify **next** path x_{n+1} with potential for **minimizing** f



Path planning

Parametrizing a navigation path $x = (x_1, \dots, x_d)$:

- $d \gg 1$ fixed **turn points**
- $M \gg 1$ discrete **actions** / turn point
 - E.g., turn angle, turn speed
 - $x_l \in \{1, \dots, M\}$
 - Engineering & communication **constraints**
- x consists of **many qualitative** factors:
 - M^d possible paths to investigate
 - **High-dimensional** experimental design



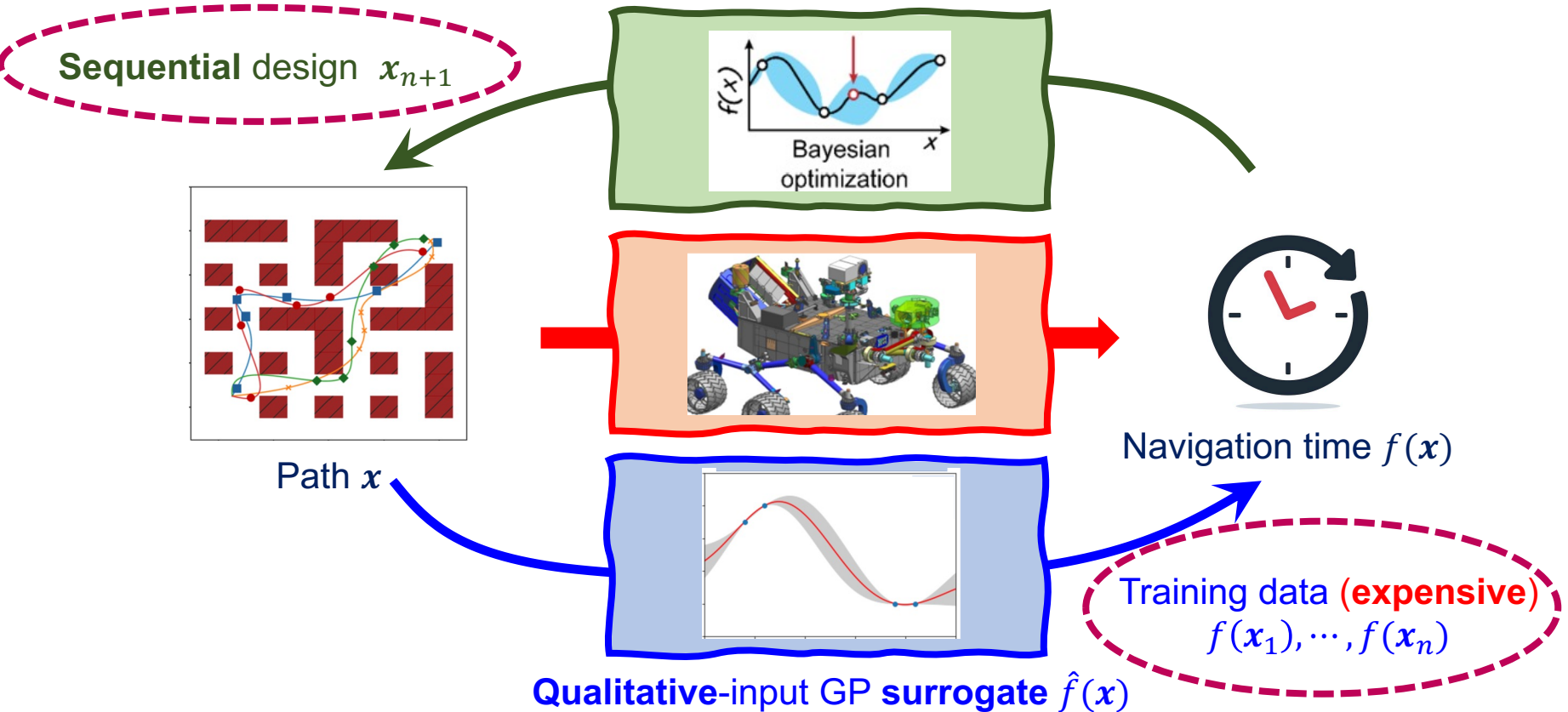
GPs for **qualitative** input factors:

- Qian et al. (2008): **exchangeable** kernel GP $f(\mathbf{x}) \sim \text{GP}(\mu, k(\cdot, \cdot))$ $\theta_l > 0$: **length-scale**

$$k(\mathbf{x}, \mathbf{x}') = \text{Cov}(f(\mathbf{x}), f(\mathbf{x}')) = \exp \left\{ - \sum_{l=1}^d \theta_l \mathbf{1}(x_l \neq x'_l) \right\}$$

- Intuition:** $\text{Cov}(f(\mathbf{x}), f(\mathbf{x}')) \downarrow$ when paths $\mathbf{x}, \mathbf{x}'$ have many **dissimilar actions**
 - Simple** model, but **effective** for experimental design
- Zhang et al. (2020): **latent variable** GP (LVGP)
 - Qualitative factors mapped to **latent continuous** space
- Roustant et al. (2020): **hypersphere kernel** GP

Experimental design



What about the **experimental design** problem?

- **Initial** experimental design
- **Sequential** experimental design

Initial design

- Question:**

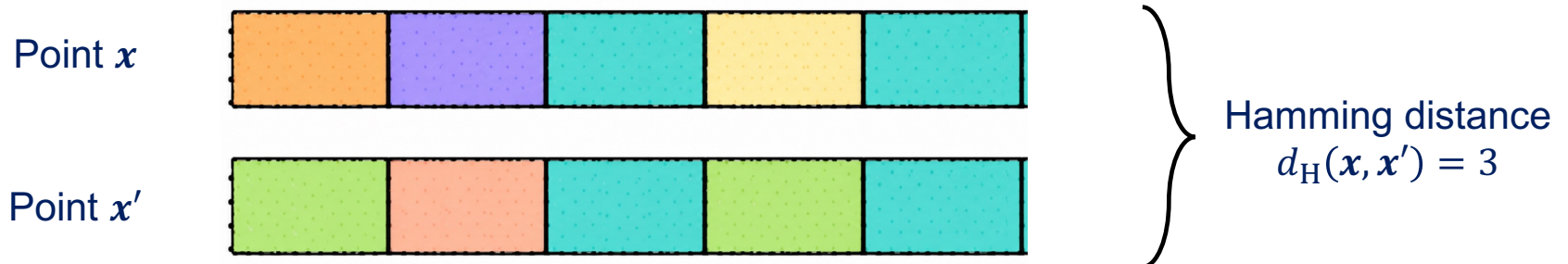
How to select **design points** $\mathcal{D} = \{x_1, \dots, x_n\}$ to maximize **learning** on f ?

- One **intuitive** design strategy:

- $d_H(x, x') = \sum_{l=1}^d \mathbf{1}(x_l \neq x'_l)$ is the **Hamming** distance
- Find the **Hamming-optimal** design $\mathcal{D}^*$ that **maximizes**:

$$\min_{x \neq x'} d_H(x, x')$$

- Intuition:** ensures **no** two points are **too close** (in Hamming distance)



Theorem: Assume $f \sim \text{GP}(\mu, k)$ follows the **exchangeable** kernel GP. Then the **Hamming-optimal** design $\mathcal{D}^*$ is **asymptotically D-optimal** for sampling f as the **length-scales** of the GP model $\theta \rightarrow \infty$. That is, $\mathcal{D}^*$ satisfies

$$\lim_{\theta \rightarrow \infty} \frac{1 - \det\{\Gamma_{\theta}(\mathcal{D}_{n,\theta}^*)\}}{1 - \det\{\Gamma_{\theta}(\mathcal{D}_n^*)\}} = 1,$$

where $\mathcal{D}_{n,\theta}^*$ is the D-optimal design under of the exchangeable GP with length-scale θ .

- This theory **bridges the gap** between **model-free designs** and **model-based optimal designs**!
 - **Model-free design**: robust to model-misspecification
 - **Model-based design**: optimal when model is correctly-specified
- Even better – the Hamming-optimal design can be formulated as an **assignment problem**!

Assignment problem (AP)

- Well-known **integer programming** problem with a rich history
- Aims to optimally **assign** workers to jobs such that total **utility** is **maximized**
- Many **generalizations**, e.g., **quadratic** assignment problem (QAP)

Formulation:

$$\max_I \sum_{i,i',j,j'} I_{ij} I_{i',j'} u_{i'j} u_{ij'}$$

$$\text{s. t. } \sum_i I_{ij} = 1 \text{ and } \sum_j I_{ij} = 1,$$

- Binary** decision variables $I_{ij} \in \{0,1\}$
 - 1 if worker i **assigned** to job j

	Job1	Job2	Job3	Job4
Worker 1	2	0	3	5
Worker 2	5	0	7	5
Worker 3	8	1	0	1
Worker 4	3	0	4	11

Assignment problem (AP)

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- Aims to optimally **assign** workers to jobs such that total **utility** is **maximized**
- Many **generalizations**, e.g., **quadratic** assignment problem (QAP)

Advantages of AP:

1. Guaranteed convergence to **global optimum**
2. Providing **optimization gaps** (not available in heuristic algorithms)
3. Can be solved efficiently by state-of-art IP optimizers (e.g., Gurobi)

- We first convert a qualitative design to dummy variables.
- For example, consider an action set $= \{a, b, c, d\}$ and design $\mathbf{x} = (b, c, d, a, d, a)$. Then,

$$I(b, c, a, d, a) = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix} & \end{matrix}.$$

- Each $I(\mathbf{x})$ must satisfy $\sum_{j=1}^m I(\mathbf{x})_{i,j} = 1$ for all i .
- The Hamming distance between designs can be represented as

$$d_H(\mathbf{x}, \mathbf{x}') = d - \text{tr}\{I(\mathbf{x})I(\mathbf{x}')^\top\}.$$

- Thus, we can represent a Hamming-optimal design problem as a **generalized AP problem**:

$$\begin{aligned} & \arg \max_{\mathcal{I}} \min_{i, i' \text{ with } i \neq i'} \{d - \text{tr}(\mathbf{I}_i \mathbf{I}_{i'}^\top)\} \\ \text{s.t. } & \sum_{k=1}^M I_{ijk} = 1, \quad \text{for all } i = 1, \dots, n \text{ and } j = 1, \dots, d, \\ & I_{ijk} \in \{0, 1\}, \quad \text{for all } i, j \text{ and } k, \end{aligned}$$

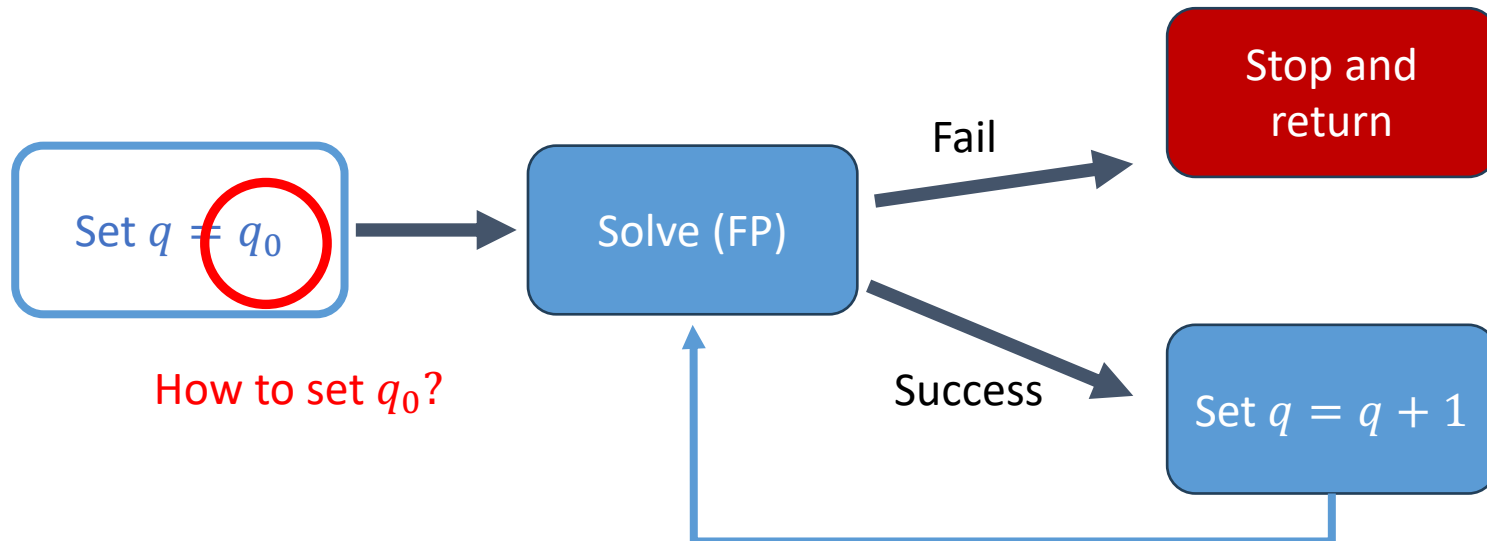
where $\mathbf{I}_i = [I_{ijk}]_{j=1}^d_{k=1}^M$ is the i –slice of the tensor $\mathcal{I}$ over its first dimension.

- It is a non-convex optimization problem involving $nd(M - 1)$ binary variables!

Optimization algorithm

We propose an iterative feasibility program (FP; Bertsimas and Tsitsiklis, 1997) to further speed up optimization time:

$$\begin{aligned} \arg \max_{\mathcal{I}} \quad & 1 \quad \text{s.t.} \quad d - \text{tr}(\mathbf{I}_i \mathbf{I}_{i'}^\top) \geq q, \quad \text{for all } i, i' = 1, \dots, n \text{ with } i \neq i', \\ & \sum_{k=1}^M I_{ijk} = 1, \quad \text{for all } i = 1, \dots, n \text{ and } j = 1, \dots, d, \\ & I_{ijk} \in \{0, 1\}, \quad \text{for all } i, j \text{ and } k. \end{aligned}$$



Lower bound for FP

Proposition: Consider the feasibility program (FP). Define the bound:

$$q_0(n, d, M) = \max \left\{ k \in \{1, \dots, d\} : \frac{M^d}{\sum_{l=0}^{k-1} \binom{d}{l} D_l} \geq n \right\}.$$

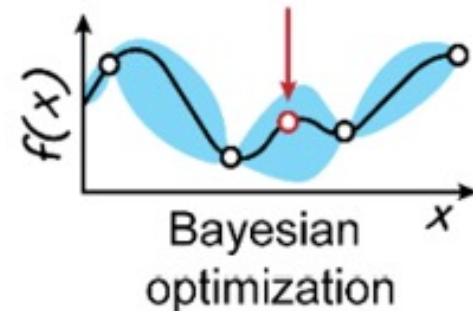
Then, for any choice of $q \leq q_0(n, d, M)$, the FP has at least one solution.

Proof: by the Gilbert-Varshamov bound (Gilbert, 1952; Varshamov, 1957)

- **Goal:** Identify **next** point with potential for **minimizing** f (i.e., navigation time)
 - **Bayesian** optimization (Frazier et al. 2018, Chen et al. 2024 *JASA*)
- **Lower confidence bound** (LCB; Auer et al. 2002):

$$x_{n+1} \leftarrow \arg \min_x \hat{f}(x) - \lambda s(x)$$

- **Exploitation:** find x with small **predicted** $\hat{f}(x)$
- **Exploration:** find x with high predictive **uncertainty** $s(x)$



Sequential design

Proposition: Assume $f \sim$ **exchangeable** kernel GP . Then the **LCB sequential design** can be formulated as a **quadratic assignment problem**.

Suppose the LCB solution is x_{n+1}^* . Its one-hot encoding $\mathbf{I}(x_{n+1}^*)$ is the optimal solution to the assignment problem with decision variables $\mathbf{I} = [\mathbf{I}_{jk}]_{j=1}^d_{k=1}^M$.

$$\arg \max_{\mathbf{I}} \sum_{\mathbf{k}=(k_1, \dots, k_d) \in [M]^d} \tilde{\xi}_{\mathbf{k}} \prod_{l=1}^d I_{lk_l} \quad \text{s.t.} \quad \sum_{k=1}^M I_{jk} = 1, \quad \text{for all } j = 1, \dots, d,$$

$$I_{jk} \in \{0, 1\}, \quad \text{for all } j \text{ and } k,$$

Advantages:

1. Guaranteed convergence to **global optimum**
2. Providing **optimization gaps** (not available in heuristic algorithms)
3. Can be solved efficiently by state-of-art IP optimizers (e.g., Gurobi)

$r=1 \ s=1$

$t \in \{r, s\} \ u=1$

$v=1$

Setup:

- Initial design size $n = 50$
- Number of qualitative levels: $M = 5, 10, 20, 30$
- Number of qualitative variables: $d = M + 15, M + 25$

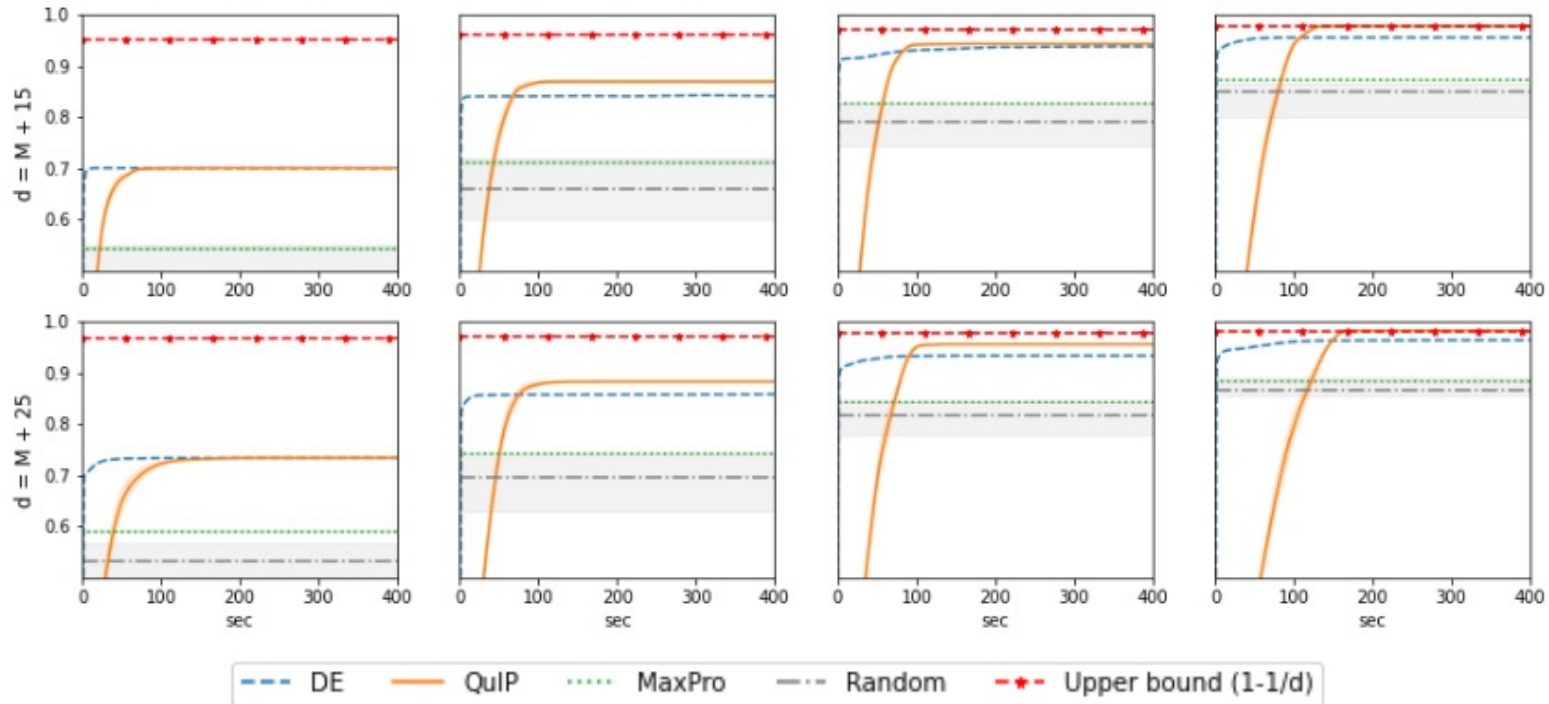
Benchmarks:

- **Random** design
- **Differential Evolution (DE)** – a metaheuristics algorithm (Stokes et al., 2023)
- **MaxPro** design (Joseph et al., 2020)
- **QuIP** - our method (Hamming-optimal design)

Metrics: minimum pairwise distance by optimization time (sec)

Simulation – Initial design

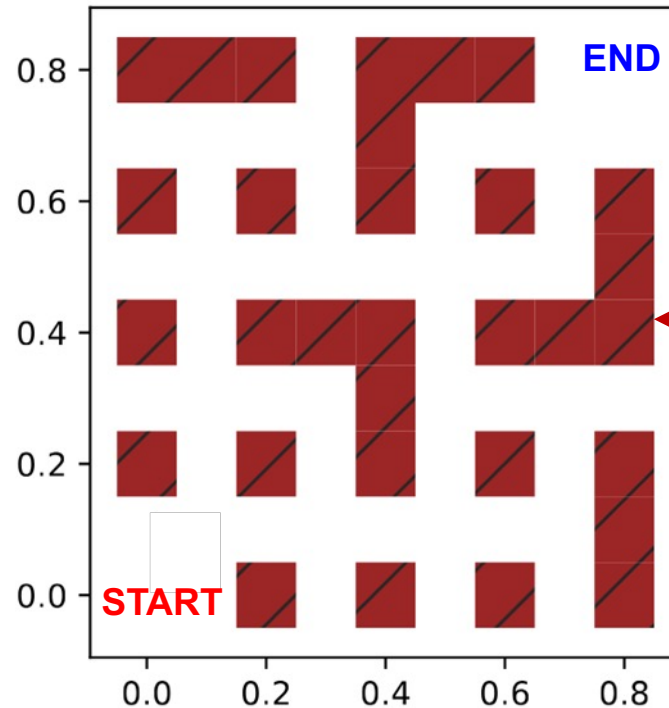
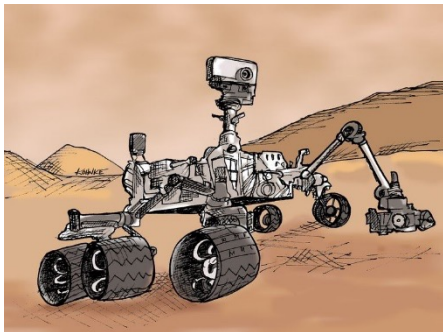
The larger (the minimum pairwise distance) is, the better.



- **Heuristics-based** algorithms tend to **stuck at local optimum**
- **QuIP (our method)** is able to achieve **better maximin designs** in almost all scenarios **with shorter time**.

Case study: Path planning

Path planning case study:

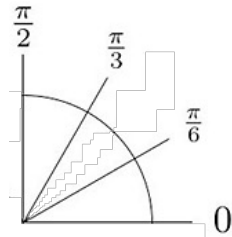


Swamps
(can pass through with greatly reduced speed)

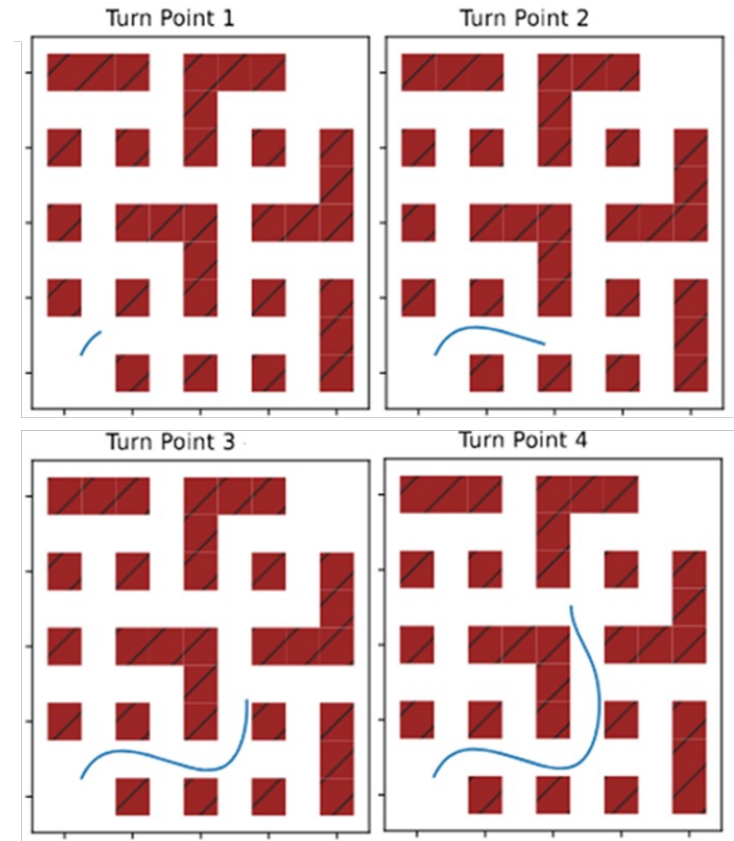
Case study: Path planning

Path planning case study:

- $d = 8$ path **turn points**
- $M = 9$ navigation **decisions** at a turn point:
 - Two **speed** choices (low, high)
 - Four **angle** choices (from $0 - \frac{\pi}{2}$ rad.)



- ... or simply **stay put**
- $9^8 \approx 43$ mil. paths to **investigate!**



Case study: Path planning

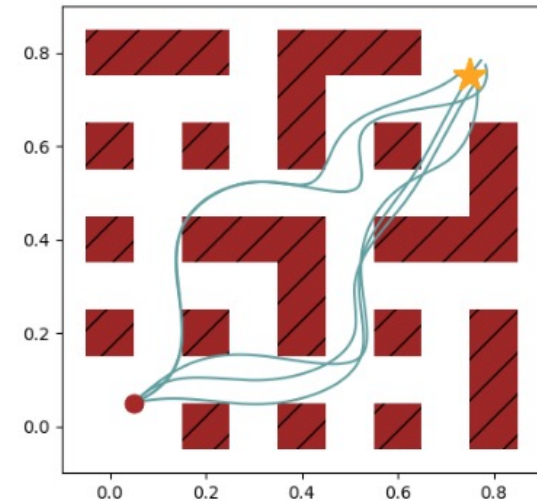
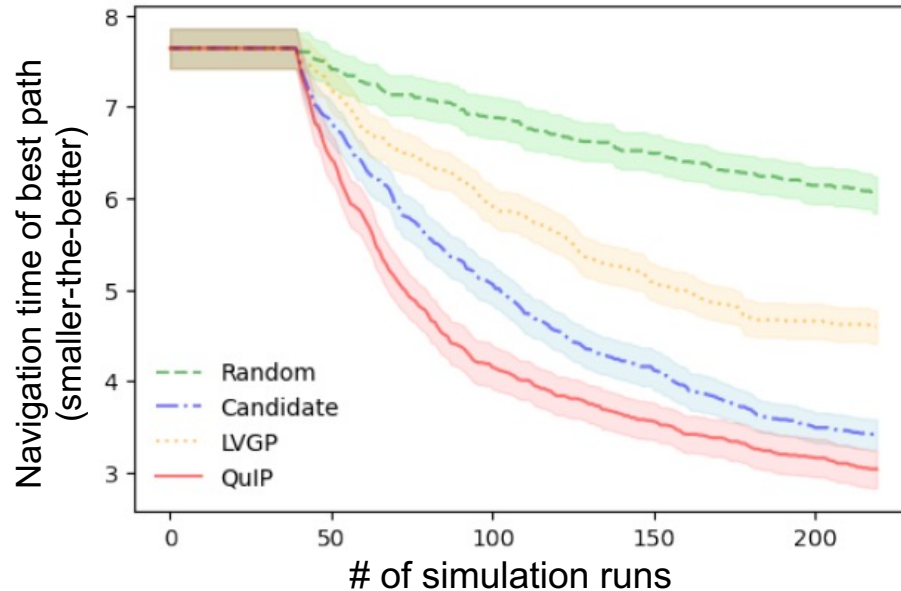
- **Simulator**: Python path planning digital twin (Wang et al. 2018)
- Proposed **QulP** approach (**exchangeable** kernel GP):
 - **Initial** design (Hamming-optimal): $n = 30$ points
 - **Sequential** design (LCB): $n_{\text{seq}} = 200$ points
 - Designs **globally** optimized via Gurobi

Benchmarks:

- **Random** design points
- **Candidate: Same** set-up as QulP, but designs optimized via random **candidate set**
- **LVGP**: Latent variable GP (Zhang et al. 2020)
 - More **complex** model, design optimization via random **candidate set**

Case study: Path planning

Path planning case study:



Examples of low cost paths discovered by QuIP

- **QuIP** performs **better** than **candidate** approach
- **LVGP** (using more **complex** model) performs **worse** than **QuIP**
 - Importance of a **well-optimized** design under limited budget

Thank you!

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